

Output

In parent class
In child class
0

~~Output~~

~~In parent class~~

~~In child class~~

~~In class A~~

~~In class B~~

~~In class C~~

~~Kanchan~~

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Aim:- Write program to understand type of inheritance.

Types of Inheritance -

1. Single Inheritance:- When one class inherits from the only one parent class, it is called inheritance.

[class A]



[class B]

2. Program - Write a program in Java to implement single inheritance.

class a {

int x;

void xyz() {

System.out.println("In parent class");

}

}

class b extends a {

void abc() {

System.out.println("In child class");

}

class Single inheritance {

public static void main(String[] args) {

b obj = new b();

obj.xyz();

obj.abc();

System.out.println(obj.x);

Teacher's Signature _____

output

In parent class

In child class

0

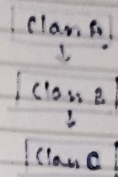
In class A

In class B

In class C

Karshan

2. Multilevel inheritance:- When a class is derived from another derived class, it is called multilevel inheritance.

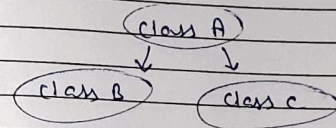


Program - write a program in Java to implement multilevel inheritance.

```
class a {
    String name = "Karshan";
    void xyz() {
        System.out.println("In class a"); } }
class b extends a {
    void abc() {
        System.out.println("In class b"); } }
class c extends b {
    void pqr() {
        System.out.println("In class c"); } }
class multilevel {
    public static void main (String[] args) {
        obj = new c ();
        obj.abc();
        obj.pqr();
        System.out.println(obj.name); } }
```


3. Multiple Inheritance:- When one class inherit from more than one class, it is called multiple inheritance.
 "It is not supported in Java".

4. Hierarchical Inheritance:- When multiple classes inherit from a single parent class, it is ~~def~~ called hierarchical inheritance.



Program Write a program in Java to implement hierarchical inheritance

```

class a {
    int x = 5;
    void b() {
        System.out.println("In class a"); } }
class b extends a {
    void b() {
        System.out.println("In class b"); }
}
class c extends a {
    void c() {
        System.out.println("In class c"); }
}
  
```

output

In class a

In class b

5

In class a

In class c

5

```

class hierarchical {
    public static void main (String[] args) {
        b obj = new b();
        obj.a();
        obj.b();
        System.out.println ((obj.x));
        c obj1 = new c();
        obj1.a();
        obj1.c();
        System.out.println ((obj1.x));
    }
}

```

5 Hybrid Inheritance:- Combination of two or more types of inheritance is called Hybrid inheritance.

Program

```

class a {
    int x = 4;
    void a() {
        System.out.println (" In class a ");
    }
}

```

```

class b extends a {
    void b() {
        System.out.println (" In class b ");
    }
}

```

```

class c extends a {
    void c() {
        System.out.println (" In class c ");
    }
}

```

Teacher's Signature _____

Output

In class a

In class c

In class d

+

In class b

In class b

+

```
class d extends c {  
    void d() {  
        System.out.println("In class d");  
    }  
}
```

```
class Hybrid {  
    public static void main (String[] args) {  
        d obj = new d();  
        obj.a();  
        obj.c();  
        obj.d();  
        System.out.println((obj));  
        b obj1 = new b();  
        obj1.a();  
        obj1.b();  
        System.out.println((obj1));  
    }  
}
```

Viva Questions

Q1. What is inheritance in Java?

Ans. Inheritance is an OOP feature in which one class requires the properties & methods of another class using the extends keyword.

Q2. What are the types of inheritance in Java?

Ans. The types of inheritance in Java are:-

- Single inheritance
- Multiple inheritance
- Hierarchical inheritance

- Multiple inheritance (through inheritances)
- Hybrid inheritance (through interfaces)

Q3 What is single inheritance?

Single inheritance occurs when one child class inherits properties from one parent class.

Ex =

```
class A {}
```

```
class B extends A {}
```

Q4 Why does java not support multiple inheritance using classes?

Java does not support multiple inheritance using classes to avoid ambiguity problems (diamond problem). It supports it through interfaces.

Q5 What is multilevel inheritance?

Multilevel inheritance occurs when one class inherits another class, and a third class inherits the second class.

Ex =

```
class A {}
```

```
class B extends A {}
```

```
class C extends B {}
```


Aim:- To study & develop applications using file handling in Java, including file creation, reading, writing & updating operation.

Theory:- File handling in Java is used to store, retrieve & manipulate data in files. It allows programs to perform operations such as creating, reading & updating files, which is essential for data persistence.

File Handling Concept:- In Java, file Handling is performed using classes from the `java.io` package. These classes provide methods to work with files & input/output streams.

File class:- The file class is used to create, delete & check properties of a file.

Examples

```
import java.io.*;
File f = new File("test.txt");
```

File Operation -

1. Create a File → Using `createNewFile()` method
2. Write to a File → Using `FileWriter` or `BufferedWriter`
3. Read from a File → Using `FileReader` or `BufferedReader`
4. Delete a file → Using `delete()` method.

output

Welcome to Java I/O
This is Output Example.

output

Enter your Name: Rahul
Enter your age: 20
Name: Rahul
Age: 20

Streams in Java:- Streams are used for input & output operations.
Input stream → reads data from file
Output stream → writes data from file

Program 1: To Display output using System.out

```
public class OutputExample {
    public static void main(String[] args) {
        System.out.println("Welcome to Java I/O");
        System.out.print("This is");
        System.out.print("Output Example");
    }
}
```

Program 2: To take input from user & display it.

```
import java.util.Scanner;
```

```
public class UserInputExample {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your Name:");
        String name = sc.nextLine();

        System.out.print("Enter your Age:");
        int age = sc.nextInt();
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        sc.close();
    }
}
```


Output

Enter first number: 10

Enter second number: 5

Sum = 15

Difference = 5

Product = 50

Division = 2

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Program 3 To take two numbers as input & perform arithmetic operation.

```
import java.util.Scanner;
```

```
public class CalculationInput {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter first number");  
        int a = sc.nextInt();
```

```
        System.out.print("Enter second number");  
        int b = sc.nextInt();
```

```
        System.out.print("Sum = " + (a+b));  
        System.out.print("Difference = " + (a-b));  
        System.out.print("Product = " + (a*b));  
        System.out.print("Division = " + (a/b));
```

```
        sc.close();  
    }  
}
```

Teacher's Signature _____

Java Questions:-

1. What is file handling in Java?

File handling in Java is used to create, read, write, update & delete files. It helps programs store data permanently instead of keeping it temporarily in memory.

2. Which package is used for file handling in Java?

The `java.io` package is mainly used for file handling in Java. It provides classes like `File`, `FileHeader`, `FileWriter`, & `BufferedReader` for file operations.

3. What is the purpose of the `File` class in Java?

The `File` class is used to create files & directories, check file information, rename files, & perform operations related to file management in Java applications.

4. What is the difference between `FileReader` & `FileWriter`?

`FileReader` & `FileWriter` are classes available in the `java.io` package used for character-based file handling in Java.

`FileReader` - used to read data from file characters by character.

`FileWriter` - used to write data into a file.

Why is exception handling important in File handling?
Exception handling is important in the file handling because file operations may generate runtime errors such as file not found, invalid file path, permission denied, or input/output errors.